CM0832 - MT5001 Elements of Set Theory
3.1 Introduction to Natural Numbers

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Semester 2026-1

Outline

Preliminaries

Textbook

Karel Hrbacek and Thomas Jech ([1978] 1999). Introduction to Set Theory.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Defining the Natural Numbers

Approaches for introducing mathematical objects

- Axiomatic
- Definitional

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Definitional approach for introducing natural numbers

- We shall define natural numbers in terms of sets.
- We shall prove the properties of natural numbers from properties of sets.

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- We shall define natural numbers in terms of sets.
- We shall prove the properties of natural numbers from properties of sets.

Question

How to define natural numbers in terms of sets?

Naive Definition

Naive idea

A natural number is the set of all smaller natural

$$
\begin{aligned}
0 &\stackrel{\text{def}}{=} \emptyset, \\
1 &\stackrel{\text{def}}{=} \{0\} &&= \{\emptyset\}, \\
2 &\stackrel{\text{def}}{=} \{0, 1\} &&= \{\emptyset, \{\emptyset\}\}, \\
3 &\stackrel{\text{def}}{=} \{0, 1, 2\} &&= \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}, \\
&\vdots
\end{aligned}
$$

Naive Defintion

A wrong impredicative definition

$$n \stackrel{\text{def}}{=} \{0, 1, \ldots, n-1\}.$$

> *"We cannot just say that a set* n *is a natural number if its elements are all the smaller natural numbers, because such a 'definition' would involve the very concept being defined." (p. 40)*

Outline

Inductive Sets

Definition [3.]1.1

Let a be a set. The **successor** of a is

$$\mathrm{S}(a) \stackrel{\text{def}}{=} a \cup \{a\}.$$

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Notation

$a + 1 \stackrel{\text{def}}{=} \mathrm{S}(a)$.

Inductive Sets

Example

$$
\begin{aligned}
0 &\stackrel{\text{def}}{=} \emptyset, \\
1 &\stackrel{\text{def}}{=} S(\emptyset) &&= S(0) = 0 + 1, \\
2 &\stackrel{\text{def}}{=} S(S(\emptyset)) &&= S(1) = 1 + 1, \\
3 &\stackrel{\text{def}}{=} S(S(S(\emptyset))) &&= S(2) = 2 + 1, \\
&\ \vdots
\end{aligned}
$$

Inductive Sets

Definition [3.]1.2

A set I is **inductive** if

(a) $\emptyset \in I$ and

(b) if $n \in I$ then $n+1 \in I$.

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(b) if $n \in I$ then $n + 1 \in I$.

Remark

An inductive set will be an infinite set.

The Set of Natural Numbers

Definition [3.]1.3

The set of **all natural numbers**, denoted $\mathbf{N}$, is defined by

$$\mathbf{N} \stackrel{\text{def}}{=} \{\, x \mid x \in I \text{ for every inductive set } I \,\}.$$

The Set of Natural Numbers

Question

Is $\mathbb{N}$ a set?

The Set of Natural Numbers

Question

Is $\mathbf{N}$ a set?

Answer

(partial) Let A be an inductive set. Using the Axiom Scheme of Comprehension we can defined the set of all natural numbers by

$$\mathbf{N} \stackrel{\text{def}}{=} \{\, x \in A \mid x \in I \text{ for every inductive set } I \,\}.$$

The Set of Natural Numbers

Question

Are there inductive sets?

The Set of Natural Numbers

Question

Are there inductive sets?

Remark

So far we only have the set $\emptyset$ and the axioms have the form: For every set X, there exists a set Y such that ....

Outline

The Axiom of Infinity

The Axiom of Existence

- *"An inductive set exits" (p. 41)*
- *"There exists an inductive set" (Enderton 1977)*
- $\exists(A)[\,\emptyset \in A \wedge \forall(a)(a \in A \rightarrow \mathrm{S}(a) \in A)\,].$

Defining Natural Numbers as Sets

Remark

So far, we defined natural numbers on terms of sets. A different point of view is stated by some authors (see, e.g. Benacerraf (1965)).

Ordering on Natural Numbers

Definition [3.]1.5

We define the relation $<$ on $\mathbf{N}$ by

$$m < n \quad \text{if and only if} \quad m \in n.$$

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Theorem

The pair $(\mathbf{N}, <)$ is a linearly ordered set.

Outline

References

Paul Benacerraf (1965). What Numbers Could not Be. The Philosophical Review 74.1, pp. 47–73. DOI: 10.2307/2183530 (cit. on p. 22).

Herbert B. Enderton (1977). Elements of Set Theory. Academic Press (cit. on p. 21).

Karel Hrbacek and Thomas Jech [1978] (1999). Introduction to Set Theory. Third Edition, Revised and Expanded. Marcel Dekker (cit. on p. 3).